

A Seminar Report
on
AUTOMATIC IMAGE CAPTIONING

**Submitted in partial fulfillment of requirements
for the award of the degree of**

**BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE & ENGINEERING
(DATA SCIENCE)**

Submitted by:

Dandu Indu mahela (22091A3243)

Under the Guidance of

Supervisor

Dr. B. Bhaskara Rao M. Tech, Ph. D
Associate Professor & HOD, Dept. of CSE(DS)

Co-Supervisor

Ms. R. Arshiya M. Tech
Assistant Professor, Dept. of CSE(DS)



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(DATA SCIENCE)**

**RAJEEV GANDHI MEMORIAL COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)**

**Affiliated to JNTUA Anantapuramu, APPROVED BY A.I.C.T.E., NEW DELHI,
ACCREDITED BY NAAC of UGC, NEWDELHI with “A+” grade,
ACCREDITED BY N.B.A, NEWDELHI,
NANDYAL-518501, (Estd-1995)**

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Rajeev Gandhi Memorial College of Engineering & Technology
(AUTONOMOUS)

APPROVED BY A.I.C.T.E., NEW DELHI, Affiliated to JNTUA, Anantapuramu,
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NANDYAL-518501



(ESTD – 1995)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (DATASCIENCE)

CERTIFICATE

This is to certify that the report entitled “**AUTOMATIC IMAGE CAPTIONING**” is a bonafide record of the technical seminar done by **DANDU INDU MAHELA (22091A3243)** in partial fulfillment of the requirements for the award of Degree of Bachelor of Technology in **Computer Science & Engineering (Data Science)** from R.G.M. College of Engineering & Technology, Nandyal during the academic year 2025-26.

Supervisor

Dr. B. Bhaskara Rao M. Tech, Ph. D

Associate Professor & HOD

Dept. of CSE(DS)

Head of the Department

Dr. B. Bhaskara Rao M. Tech, Ph. D

Associate Professor & HOD

Dept. of CSE(DS)

Co-Supervisor

Ms. R. Arshiya M. Tech

Assistant Professor

Dept. of CSE(DS)

Place: Nandyal

Date:

Candidate's Declaration

I hereby declare that the technical seminar entitled “**Automatic image captioning**” was carried out and written by me under the guidance of **Dr. B. Bhaskara Rao** M. Tech, Ph. D., **Associate Professor & HOD**, **Ms. R. Arshiya** M. Tech, **Assistant Professor**, Dept. of Computer Science & Engineering (Data Science), R.G.M. College of Engineering & Technology, and this technical seminar report is submitted in the partial fulfillment of the requirements for the award of the degree of “Bachelor of Technology” in **Computer Science & Engineering (Data Science)**. I have not submitted the matter presented in this report anywhere for the award of any other Degree.

(DANDU INDU MAHELA)

Place: Nandyal

Date:

Roll No.: 22091A3243

IV B. Tech II Sem, CSE(DS),
RGM CET.

Preface

I have made this report file on the topic **AUTOMATIC IMAGE CAPTIONING**, I have tried my best to elucidate all the relevant detail to the topic to be included in the report. While in the beginning I have tried to give a general view about this topic.

My efforts and wholehearted cooperation of each and every one has ended on a successful note. I express my sincere gratitude to **Dr. B. Bhaskara Rao** M. Tech, Ph. D, **and Ms. R. Arshiya** M. Tech, who assisting me throughout the preparation of this topic. I thank him for providing me the reinforcement, confidence and most importantly the track for the topic whenever I needed it.

DANDU INDU MAHELA
(22091A3243)

ABSTRACT

Image captioning is an important task in the field of Artificial Intelligence that combines Computer Vision and Natural Language Processing to automatically generate textual descriptions for images. The main objective of this project is to develop an intelligent system that can understand visual content and produce meaningful captions in natural language. The proposed Image Caption Generator uses advanced deep learning techniques to achieve this task. In this system, the input image is first preprocessed and converted into a format suitable for model analysis. A Vision Transformer (ViT) is used to extract important visual features from the image by dividing it into patches and analyzing relationships between them using a self-attention mechanism. These extracted features are then passed to the GPT-2 language model, which generates a coherent textual caption describing the image content. The system is implemented using Python and integrated with a Streamlit-based web interface, which allows users to upload images and view the generated captions easily through an interactive application. In addition to caption generation, the system also provides an accuracy score that indicates the confidence level of the predicted caption, helping users understand the reliability of the result. By combining transformer-based vision models and powerful language generation models, the proposed system provides an efficient and scalable solution for automated image description. This technology has potential applications in assistive systems for visually impaired individuals, image search engines, social media content management, surveillance systems, and intelligent multimedia analysis.

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CHAPTER 1

INTRODUCTION

With the rapid growth of digital technology, a massive amount of visual data such as images and videos is generated every day through smartphones, social media, surveillance systems, and online platforms. Understanding and interpreting this visual data automatically has become a major research area in Artificial Intelligence. Image captioning is an advanced technique that combines Computer Vision and Natural Language Processing to automatically generate meaningful textual descriptions for images. The main goal of image captioning is to enable machines to understand the content of an image and express it in natural human language.

This technology has become increasingly important because it helps bridge the gap between visual information and textual understanding. Image captioning plays a crucial role in various applications such as assistive technologies for visually impaired individuals, automatic image indexing, social media content management, and intelligent search systems. By converting images into descriptive text, machines can make visual information more accessible and easier to process.

In recent years, deep learning models have significantly improved the performance of image captioning systems. Traditional approaches mainly used Convolutional Neural Networks (CNN) for image feature extraction and Recurrent Neural Networks (RNN) for sequence generation. However, modern approaches are increasingly adopting transformer-based architectures such as Vision Transformers (ViT) and GPT models, which provide better context understanding through self-attention mechanisms. These advanced models can capture complex relationships within images and generate more accurate and natural captions.

Therefore, this project focuses on developing an Image Caption Generator using Vision Transformer and GPT-2 models, along with a Streamlit-based interface that allows users to upload images and generate captions along with an accuracy score indicating the confidence of the prediction.

CHAPTER 2

LITERATURE REVIEW

Image captioning has been an active research area in Artificial Intelligence because it combines two major fields: Computer Vision, which focuses on understanding visual content, and Natural Language Processing (NLP), which deals with generating human language. The goal of image captioning is to enable machines to analyze images and automatically generate meaningful textual descriptions. Early research in this field mainly relied on template-based approaches, where detected objects in images were inserted into predefined sentence structures to form captions. However, these methods were limited because they could only produce simple and rigid descriptions and lacked the ability to generate natural and flexible sentences.

With the advancement of deep learning techniques, researchers started using neural network models to improve caption generation. Convolutional Neural Networks (CNNs) were widely used to extract important visual features from images, such as objects, shapes, and patterns. These extracted features were then passed to Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks to generate captions sequentially, word by word. Later, attention mechanisms were introduced, allowing models to focus on specific regions of an image while generating each word of the caption, which significantly improved caption quality and accuracy.

In recent years, transformer-based architectures have revolutionized the field of image captioning. Models such as Vision Transformers (ViT) for image feature extraction and GPT-based language models for text generation have demonstrated superior performance compared to traditional CNN–RNN approaches. These modern models use self-attention mechanisms to capture complex relationships within images and generate more context-aware and natural language descriptions. As a result, transformer-based image captioning systems have become widely used in applications such as assistive technologies for visually impaired individuals, intelligent image search systems, automated content management, and multimedia analysis.

Technologies Used in Existing Systems

- **Convolutional Neural Networks (CNN)** – Used for extracting visual features from images.
- **Recurrent Neural Networks (RNN)** – Used to generate captions sequentially based on image features.
- **Long Short-Term Memory (LSTM)** – An improved version of RNN that handles long-term dependencies in text generation.
- **Attention Mechanism** – Helps the model focus on important regions of an image when generating each word in the caption.
- **Transformer Models** – Advanced deep learning models that use self-attention for better context understanding.
- **Vision Transformer (ViT)** – A transformer-based model used for extracting image features.
- **GPT-based Models** – Language models used for generating human-like captions.

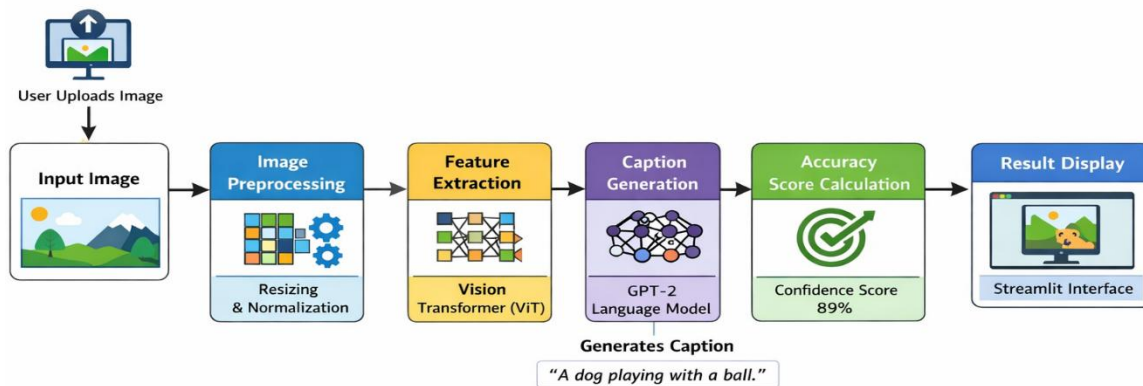
Important Developments in the Field

- **Attention-Based Captioning (Show, Attend and Tell – 2016)** – Introduced attention mechanisms to improve caption accuracy.
- **Transformer-Based Models (2017 onwards)** – Use self-attention mechanisms to improve contextual understanding.
- **Vision Transformer and GPT Integration** – Modern systems combine vision transformers with language models for better caption generation.
- **Multimodal AI Systems** – Recent developments focus on combining image and text understanding in a single framework.

CHAPTER 3

WORKING PRINCIPLES

Automatic Image Captioning

Image Caption Generator

Automatic Image Captioning

The proposed Image Caption Generator works by integrating **computer vision and natural language processing techniques** to automatically generate descriptive captions for images. The system uses deep learning models such as **Vision Transformer (ViT)** for visual feature extraction and **GPT-2** for natural language generation. The overall process involves several stages, including image acquisition, preprocessing, feature extraction, caption generation, accuracy evaluation, and result display. Each stage contributes to transforming visual information into meaningful textual descriptions.

Steps in Automatic Image Captioning

1. Image Input

The first step of the system involves obtaining an input image from the user. The application provides a **user-friendly interface built using Streamlit**, where users can upload images from their local device. The system supports common image formats such as JPG, PNG, and JPEG. Once the image is uploaded, it is temporarily stored and prepared for further processing. This stage allows the system to receive real-world visual data that will be analyzed by the deeplearning model.

2. Image Preprocessing

After receiving the image, preprocessing is performed to prepare the image for the deep learning model. Images in raw form may have different sizes, resolutions, and pixel values, which may not be suitable for direct model input. Therefore, preprocessing operations such as **resizing the image to a fixed dimension, converting it into tensor format, and normalizing pixel values** are performed. This stage ensures consistency in the input data and improves the efficiency and performance of the model. Proper preprocessing also reduces noise and helps the model focus on important visual features.

3. Feature Extraction using Vision Transformer (ViT)

Once preprocessing is completed, the image is passed to the **Vision Transformer (ViT)** model for feature extraction. Unlike traditional CNN-based approaches, the Vision Transformer divides the image into **small patches**, similar to tokens in natural language processing. Each patch is converted into a vector representation and fed into the transformer architecture. Using the **self-attention mechanism**, the model analyzes relationships between different image patches and identifies important patterns, objects, and contextual information. This process enables the model to capture both **local and global features** present in the image. The output of this stage is a high-level feature representation that summarizes the visual content of the image.

4. Caption Generation using GPT-2

The extracted visual features are then passed to the **GPT-2 language model**, which is responsible for generating natural language descriptions. GPT-2 is a transformer-based generative model that has been pre-trained on large text datasets. It interprets the visual features provided by the Vision Transformer and predicts words sequentially to form a meaningful sentence. The model uses contextual understanding and probability-based prediction to generate captions that accurately describe the objects and activities present in the image. This stage converts the visual understanding of the image into human-readable text.

5. Accuracy or Confidence Score Calculation

In addition to caption generation, the system calculates a confidence or accuracy score for the generated caption. This score is derived from the probability values generated by the language model during word prediction. By evaluating the likelihood of predicted words, the system estimates how reliable the generated caption is. The accuracy score helps users understand the confidence level of the model's prediction, providing additional information about the quality of the caption.

6. Displaying the Output

In the final stage, the generated caption and the corresponding accuracy score are displayed to the user through the Streamlit web interface. The interface shows the uploaded image along with the predicted caption and confidence value. This allows users to easily interpret the results and interact with the system. The combination of image visualization, caption output, and accuracy measurement makes the system more informative and user-friendly.

CHAPTER 4

IMPLEMENTATION / UTILIZATION

Overview

The Image Caption Generator system is implemented using a combination of Computer Vision and Natural Language Processing techniques. The goal of the system is to automatically generate a meaningful textual description for a given image. This is achieved by using a deep learning architecture that connects an image understanding model with a language generation model.

In this project, the system is implemented using Python and several advanced AI libraries such as PyTorch and Hugging Face Transformers. A pretrained VisionEncoderDecoderModel is used, which combines a Vision Transformer (ViT) encoder for extracting visual information from the image and a language model decoder for generating descriptive text.

The project also includes a Streamlit-based web interface that allows users to upload an image and generate captions interactively. Once the user uploads an image, the system preprocesses the image using ViTImageProcessor, converts it into tensor format, and sends it to the deep learning model. The model analyzes the visual content and generates a caption describing the objects and activities in the image.

The output caption is then decoded into human-readable text using a tokenizer, and the final result is displayed on the interface along with the uploaded image.

This implementation demonstrates how deep learning models can integrate visual perception and natural language generation to create intelligent systems capable of describing images automatically.

Tools / Platform Used

1.Python

Python is the main programming language used to develop the Image Caption Generator system. It provides powerful libraries for machine learning, deep learning, and image processing. Python also makes it easier to integrate AI models and build applications efficiently.

2.PyTorch

PyTorch is an open-source deep learning framework used to run neural network models. It helps in processing image data as tensors and performing model inference. In this project, PyTorch is used to load and execute the pretrained image captioning model.

3.Transformers

Transformers is a library developed by Hugging Face that provides pretrained models for natural language processing and vision tasks. It simplifies the implementation of complex AI models. In this project, it is used to load the image captioning architecture and generate captions.

4.Streamlit

Streamlit is a Python framework used to create interactive web applications for machine learning projects. It allows users to upload images and view generated captions through a simple interface. It helps in demonstrating the working of the system visually.

5.Pillow

Pillow is a Python library used for image processing and manipulation. It allows the program to open, read, and modify image files. In this project, it is used to load the uploaded image before sending it to the model.

6.Tokenizer

Tokenizer is used to convert text into tokens and decode tokens back into readable sentences. It helps the model understand and generate language efficiently. In this project, it converts the model output into a human-readable caption.

7. VisionEncoderDecoderModel

VisionEncoderDecoderModel is a deep learning architecture that combines an image encoder and a text decoder. The encoder extracts features from the image, while the decoder generates a descriptive caption. This model enables the system to convert visual information into natural language.

8. ViTImageProcessor

ViTImageProcessor is used to preprocess images before they are fed into the deep learning model. It resizes, normalizes, and converts images into tensor format. This ensures that the image is in the correct format required by the Vision Transformer encoder.

- **Dataset Used**

The Image Caption Generator in this project uses a pretrained model from the Transformers library. This model was originally trained on the MS COCO (Microsoft Common Objects in Context) dataset, which is a widely used dataset for image captioning and computer vision tasks.

About the MS COCO Dataset

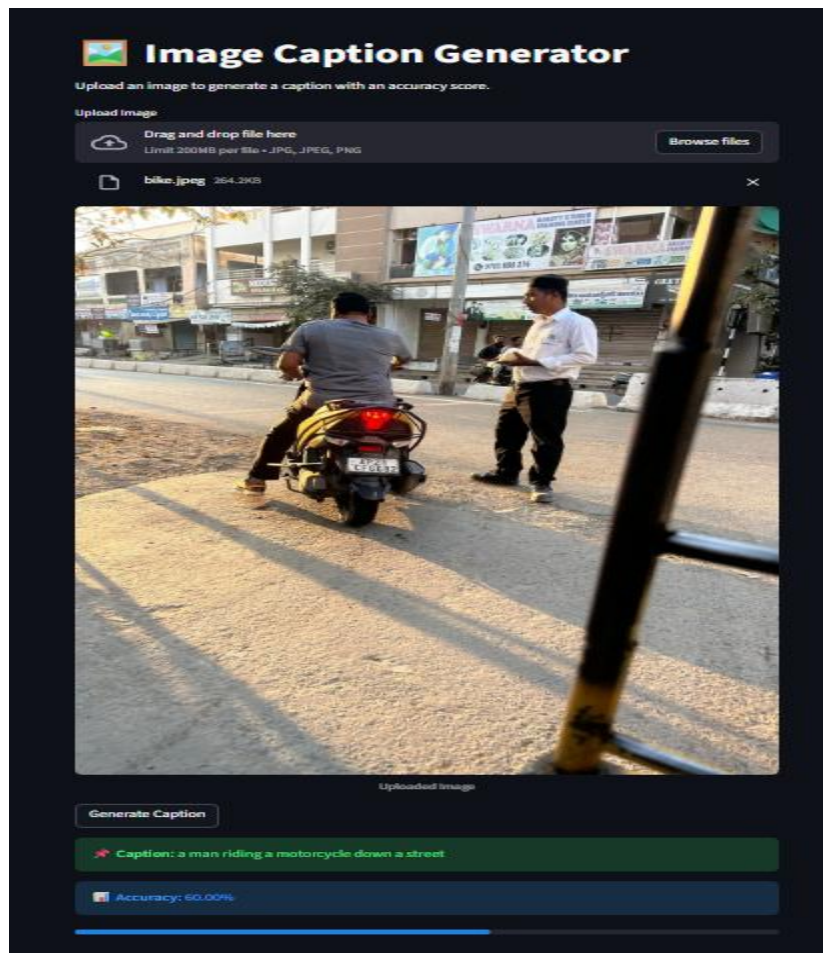
The MS COCO dataset contains a large collection of real-world images showing everyday scenes such as people, animals, vehicles, and objects. Each image in the dataset is provided with multiple captions written by humans that describe the objects and actions in the image.

Important features of the dataset include:

- More than 330,000 images
- Around 1.5 million captions
- About 80 object categories
- Each image has multiple human-written descriptions

These captions help the model learn how to describe images in natural language.

Output



CHAPTER 5

APPLICATIONS

1. Assistance for Visually Impaired People

Image captioning systems are very useful for people who are visually impaired or blind. These systems automatically generate textual descriptions of images by analyzing the objects and activities present in the image. The generated captions can then be converted into speech using screen reader technologies. This allows visually impaired users to understand images on websites, social media platforms, educational materials, and digital documents. For example, if an image contains a dog playing in a park, the system may generate a caption such as *“A dog running on grass in a park.”* The screen reader can read this caption aloud, helping the user understand the image content without needing assistance from others. This improves accessibility and independence for visually impaired individuals.

2. Social Media and Content Creation

Image caption generators can help users and content creators automatically generate captions for images uploaded to social media platforms such as Instagram, Facebook, Twitter, and LinkedIn. Writing captions for every image can take time, especially for users who frequently post images. By using an image captioning system, the platform can automatically analyze the image and generate a suitable description. This helps users quickly create posts and makes social media content more accessible. It is also useful for digital marketers and influencers who manage multiple social media accounts and need to post content regularly. Additionally, automatically generated captions can improve searchability and engagement by providing meaningful descriptions of visual content.

3. Image Search and Retrieval Systems

Image captioning technology improves image search engines and retrieval systems. Normally, search engines rely on manually added tags or metadata to identify images. However, this process can be time-consuming and may not always provide accurate descriptions.

With image captioning, images are automatically analyzed and converted into descriptive text. This allows search engines to index images based on the generated captions. For example, if a user searches for *“a person riding a bicycle on a road,”* the search system can retrieve images whose captions contain similar descriptions.

4. Digital Asset Management

Many organizations such as media companies, news agencies, advertising firms, and research institutions manage thousands or even millions of images. Manually adding descriptions to each image is a time-consuming task. Image caption generators can automatically create metadata and descriptions for images stored in digital asset management systems. This makes it easier to organize, categorize, and retrieve images when needed. For example, a news agency storing thousands of photographs can quickly identify images related to specific events or locations using automatically generated captions. This improves efficiency, organization, and productivity in managing digital media resources.

5. Robotics and Autonomous Systems

Image captioning can also be used in robots and autonomous systems to help machines understand their surroundings. Robots often rely on computer vision systems to detect objects and analyze the environment. Image captioning adds an additional layer by converting visual information into descriptive language.

For example, a robot in a smart home environment may generate captions such as *“A person sitting on a sofa using a laptop.”* This helps the robot interpret human activities and respond accordingly. In autonomous vehicles and surveillance systems, image captioning can help in describing road conditions, identifying objects, and understanding complex scenes. This improves decision-making and situational awareness in intelligent systems.

6. E-Commerce and Online Shopping Platforms

Online shopping platforms such as Amazon, Flipkart, and eBay contain millions of product images. Writing detailed descriptions for every product manually requires significant time and effort.

Image captioning systems can automatically generate basic product descriptions by analyzing product images. For example, an image of a shoe might generate a caption like *“A pair of black running shoes with white soles.”* These descriptions can be used as product metadata to improve product search results and help customers understand the product better. This technology also helps sellers quickly upload products without spending time writing detailed descriptions.

7.Multimedia Content Analysis

Image captioning plays an important role in multimedia content analysis, where systems automatically understand and describe visual content from images and videos. In multimedia platforms, large amounts of visual data such as images, movies, and video clips are generated every day. Manually analyzing and describing this content is difficult and time-consuming.

Image captioning systems help solve this problem by automatically generating textual descriptions for images and video frames. In video analysis, a video can be divided into multiple frames, and each frame can be processed by the image captioning model. The system then generates captions describing the objects, people, and activities present in each frame.

8.Advertisement and Marketing

In the advertising and marketing industry, companies frequently use images in promotional campaigns, digital advertisements, and social media posts. Image caption generators can automatically create descriptions for these images, which can be used in marketing content. This saves time for marketing teams and helps create engaging content quickly.

CHAPTER 6 ADVANTAGES

1. Improves Accessibility for Visually Impaired People

One of the most important advantages of image caption generators is that they help visually impaired or blind individuals understand images. The system automatically generates a text description of an image by analyzing the objects and actions present in it. This text can be read aloud using screen readers or assistive technologies. As a result, visually impaired users can understand the content of images on websites, social media, and digital documents without needing assistance from others. This improves digital accessibility and inclusivity.

2. Saves Time and Reduces Manual Work

Manually writing captions or descriptions for a large number of images requires significant time and effort. Image caption generators automate this process by analyzing images and generating captions instantly. This is very useful for organizations that manage large image collections, such as social media platforms, media companies, and e-commerce websites. Automation helps reduce manual work and allows employees to focus on more important tasks.

3. Improves Image Search and Retrieval

Image captioning converts visual information into textual descriptions, which makes it easier for search systems to understand images. These captions can be used as metadata to index images in search engines. Users can then search for images using keywords or phrases. For example, if someone searches for *“a cat sitting on a sofa,”* the system can quickly find images that contain similar descriptions. This improves the efficiency and accuracy of image search systems.

4. Helps in Managing Large Image Databases

Organizations that store large numbers of images often face challenges in organizing and managing them. Image caption generators automatically create descriptions and tags for images, which helps categorize and organize them more efficiently. This makes it easier to store, retrieve, and analyze images in digital libraries, media archives, and research databases.

5. Supports Artificial Intelligence and Automation

Image captioning systems combine computer vision and natural language processing technologies. This allows machines to understand visual information and describe it in human language. Such systems can be integrated into AI applications like robots, smart assistants, surveillance systems, and autonomous vehicles. This improves the ability of machines to interpret visual environments and interact with humans.

6. Enhances User Experience

Automatically generated captions help users quickly understand the content of an image without needing to analyze it manually. This is especially useful on social media platforms, online shopping websites, and educational platforms. Providing clear descriptions improves user engagement and helps users find relevant content more easily.

CHAPTER 7

LIMITATIONS

1. Limited Understanding of Complex Scenes

Image captioning systems sometimes struggle to understand complex images that contain many objects, activities, or interactions. While the model may detect objects correctly, it may not fully understand the relationship between them. As a result, the generated caption may be incomplete or may not fully describe the scene accurately.

2. Dependence on Training Data

The performance of an image caption generator depends heavily on the quality and size of the training dataset. If the model is trained on limited or biased data, it may not recognize certain objects, environments, or cultural contexts. This can lead to inaccurate or less meaningful captions.

3. Difficulty in Understanding Context

Image captioning models mainly focus on visual features extracted from the image. However, understanding the context or background meaning of an image can be difficult for the model. For example, the system may recognize objects in an image but may not correctly interpret the situation or activity taking place.

4. Inaccurate or Generic Captions

Sometimes the system generates very general captions such as “A person standing in a room” even if the image contains more specific details. This happens because the model may not always capture fine details or unique characteristics of the image.

5. High Computational Requirements

Training and running image captioning models often requires high computational resources, including GPUs and large amounts of memory. This can make the system expensive to develop and deploy, especially for organizations with limited resources.

CHAPTER 8

FUTURE SCORE

The future of image captioning technology is very promising as advancements in artificial intelligence, deep learning, and computer vision continue to grow. Future models will be able to generate more accurate and detailed captions by better understanding complex scenes, object relationships, and human activities within images. Improved neural network architectures and larger datasets will allow systems to produce captions that are closer to human-level descriptions. Another important area of future development is multilingual image captioning. Currently, many image captioning systems generate captions primarily in English. In the future, these systems can be enhanced to support multiple languages, allowing users from different regions to understand image content in their native language. This will improve global accessibility and make the technology more useful for international applications.

Image captioning can also be integrated with advanced technologies such as augmented reality (AR), virtual reality (VR), and robotics. For example, AR devices could provide real-time descriptions of objects in the environment, helping users understand their surroundings better. In robotics, image captioning systems can help robots interpret visual information and communicate what they see, improving interaction between humans and machines.

In addition, future image captioning systems may be combined with video analysis and real-time processing to generate captions for live video streams. This can be useful in areas such as surveillance, smart cities, healthcare monitoring, and autonomous vehicles. With continuous research and technological improvements, image captioning systems will become more intelligent, efficient, and widely used in many real-world applications.

CHAPTER 9

CONCLUSION

The Automating Image Caption Generator is an important application of artificial intelligence that combines computer vision and natural language processing to automatically generate textual descriptions for images. By analyzing visual features in an image, the system is able to identify objects, scenes, and activities, and convert this information into meaningful captions. This helps machines understand visual content and communicate it in human-readable language.

The project demonstrates how deep learning models such as Convolutional Neural Networks (CNNs) for feature extraction and Recurrent Neural Networks (RNNs) or Transformer-based models for language generation can work together to produce captions. These technologies enable the system to learn patterns from large datasets and generate captions that describe the image content effectively.

Image captioning systems have many real-world applications in areas such as assistive technologies for visually impaired people, multimedia content analysis, digital libraries, surveillance systems, and social media platforms. By automatically generating captions, these systems reduce manual effort and improve accessibility, content organization, and user experience. Overall, the Image Caption Generator project highlights the potential of artificial intelligence in understanding and interpreting visual information. With continuous advancements in machine learning and deep learning technologies, image captioning systems will become more accurate, efficient, and widely used in future applications.